

arxiv.org/abs/2504.03553

KnowSelf: Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching LLM agents WHEN to think fast, reflect slowly, or consult external knowledge — on demand.

 Fast

 Slow

 Knowledgeable

Current Agent Training Uses 'Flood Irrigation'

Indiscriminate knowledge injection hurts efficiency — and performance

① Trajectory Imitation

e.g. ReAct

Blindly copies gold trajectories.
No adaptation to agent capability at each step.

② Trial-and-Error Refinement

e.g. Reflexion

Applies reflection/feedback universally,
even when the agent already knows the
answer.

③ Knowledge-Augmented Planning

e.g. KnowAgent, WKM

Injects full domain knowledge at every step,
regardless of actual need.

→ All three share the same flaw:

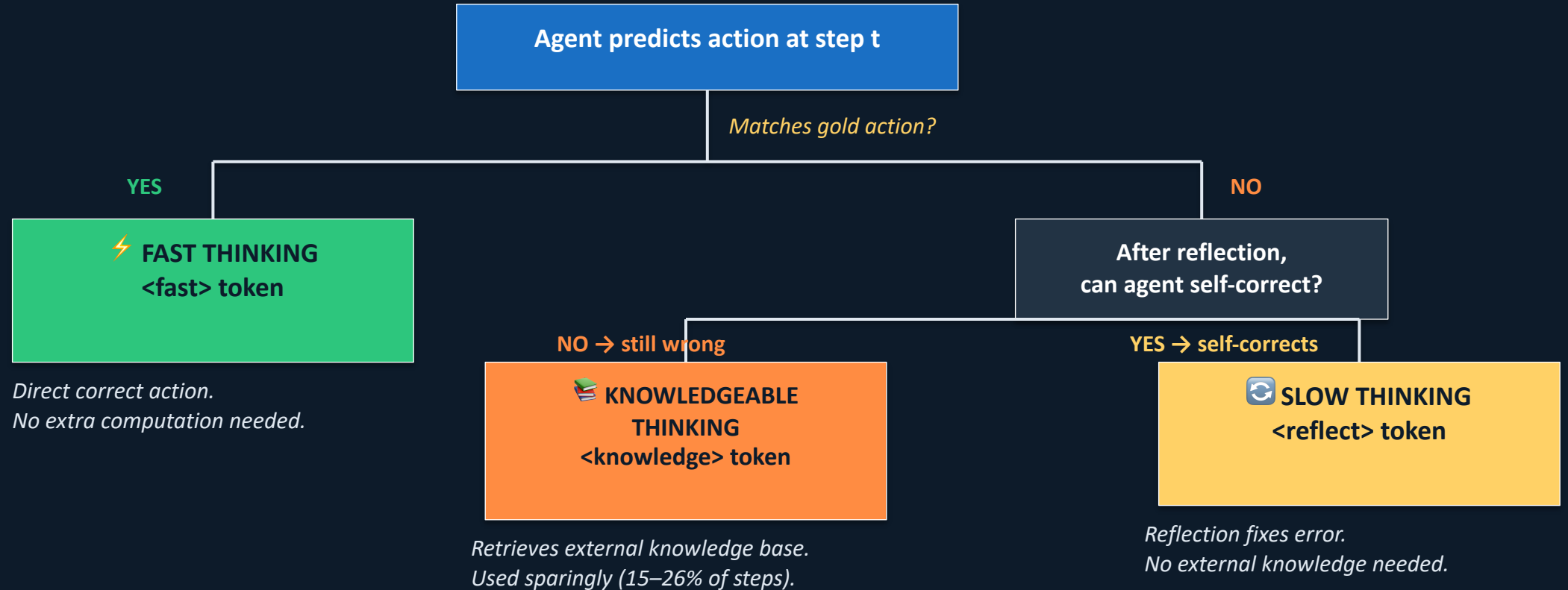
"FLOOD IRRIGATION" — inject knowledge/feedback/trajectories regardless of step-level need

Consequences:

- ✗ Fragile planning on unexpected inputs (pattern collapse)
- ✗ High inference cost from unnecessary knowledge retrieval
- ✗ Agents lack metacognitive self-regulation — unlike humans

Three Situational Modes: Fast, Slow & Knowledgeable Thinking

KnowSelf classifies each decision step with a heuristic criterion → inserts special tokens



Special tokens (<fast> / <reflect> / <knowledge>) are generated at inference — triggering the appropriate cognitive mode automatically

KnowSelf Framework: Data Construction → Two-Stage Training → Self-Aware Inference

A lightweight, three-step pipeline that teaches agents situational self-regulation

Step 1 Self-Awareness Data Construction

Agent self-explores environment.
Heuristic criterion classifies each step:

- Correct → <fast>
- Reflection fixes → <reflect>
- Neither works → <knowledge>

No human annotation required.

Step 2 Two-Stage Training

Stage A — Supervised Fine-Tuning (SFT):
Teaches initial self-awareness patterns.

Stage B — Reward-based Policy
Optimization (RPO):
Refines token generation quality.

Step 3 Self-Aware Inference

Agent generates situational token;
runtime routes to:

- ⚡ Direct action
- 🔄 Self-reflection loop
- 📖 External KB retrieval

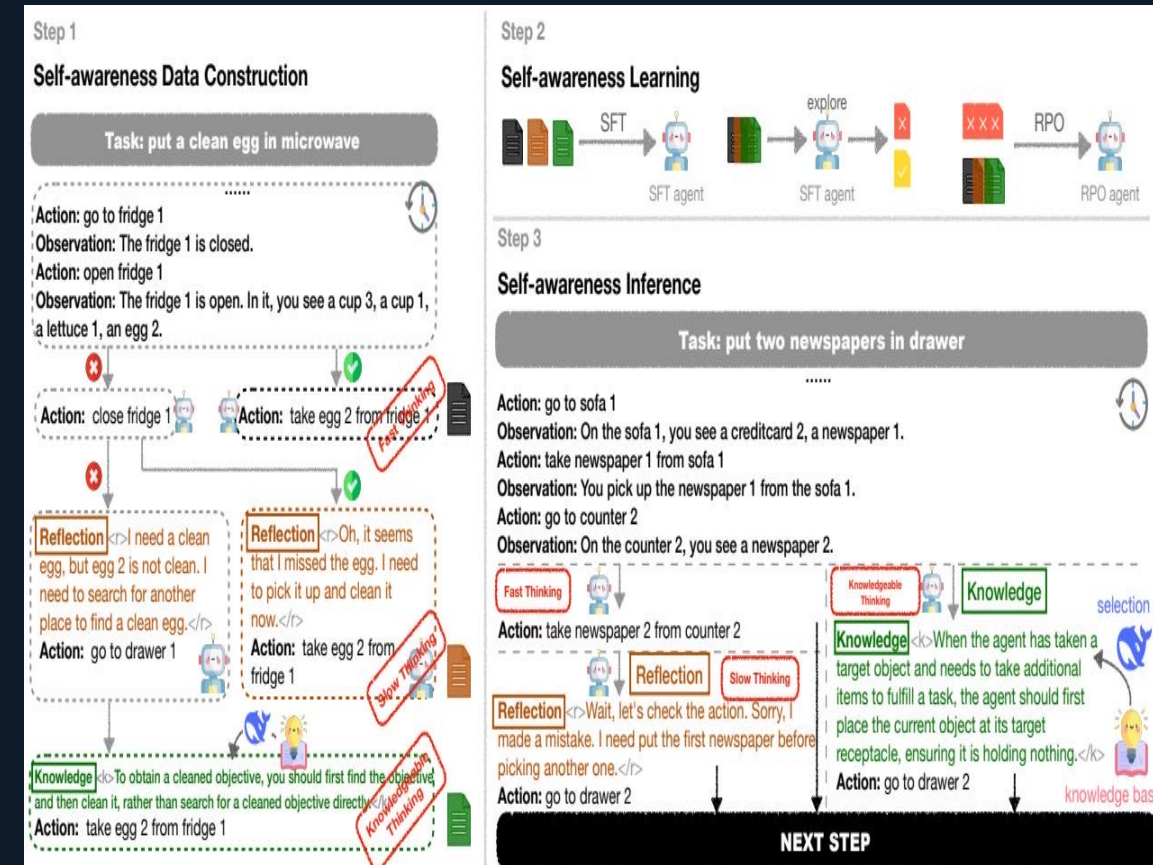


Figure: KnowSelf full pipeline (arxiv 2504.03553, Fig. 3)

KnowSelf Correctly Heats a Mug and Places a Saltshaker — Where Baselines Fail

Qualitative comparison: KnowSelf (Llama-8B) vs. OpenAI-O1 and DeepSeek-R1

Task A: "put a hot mug in cabinet"

OpenAI-O1 (baseline):

Opens microwave → tries to take EGG (irrelevant object)

✗ Incorrect — diverts to wrong object

KnowSelf (Llama-8B):

 <reflect> token

Realizes the open microwave should be used to HEAT the mug → correct sequence

✓ Correct

Task B: "put a saltshaker in drawer"

DeepSeek-R1 (baseline):

Reflects that drawer is open → tries to CLOSE it first

✗ Fails — action not required and breaks the plan

KnowSelf (Llama-8B):

 <knowledge> token

Retrieves knowledge: place object without removing unrelated items → correct placement

✓ Correct

Reference: Figure 1, arxiv 2504.03553 — qualitative comparison



KnowSelf Achieves 84.33% on ALFWorld with Llama-8B — Using Only 15% Knowledge

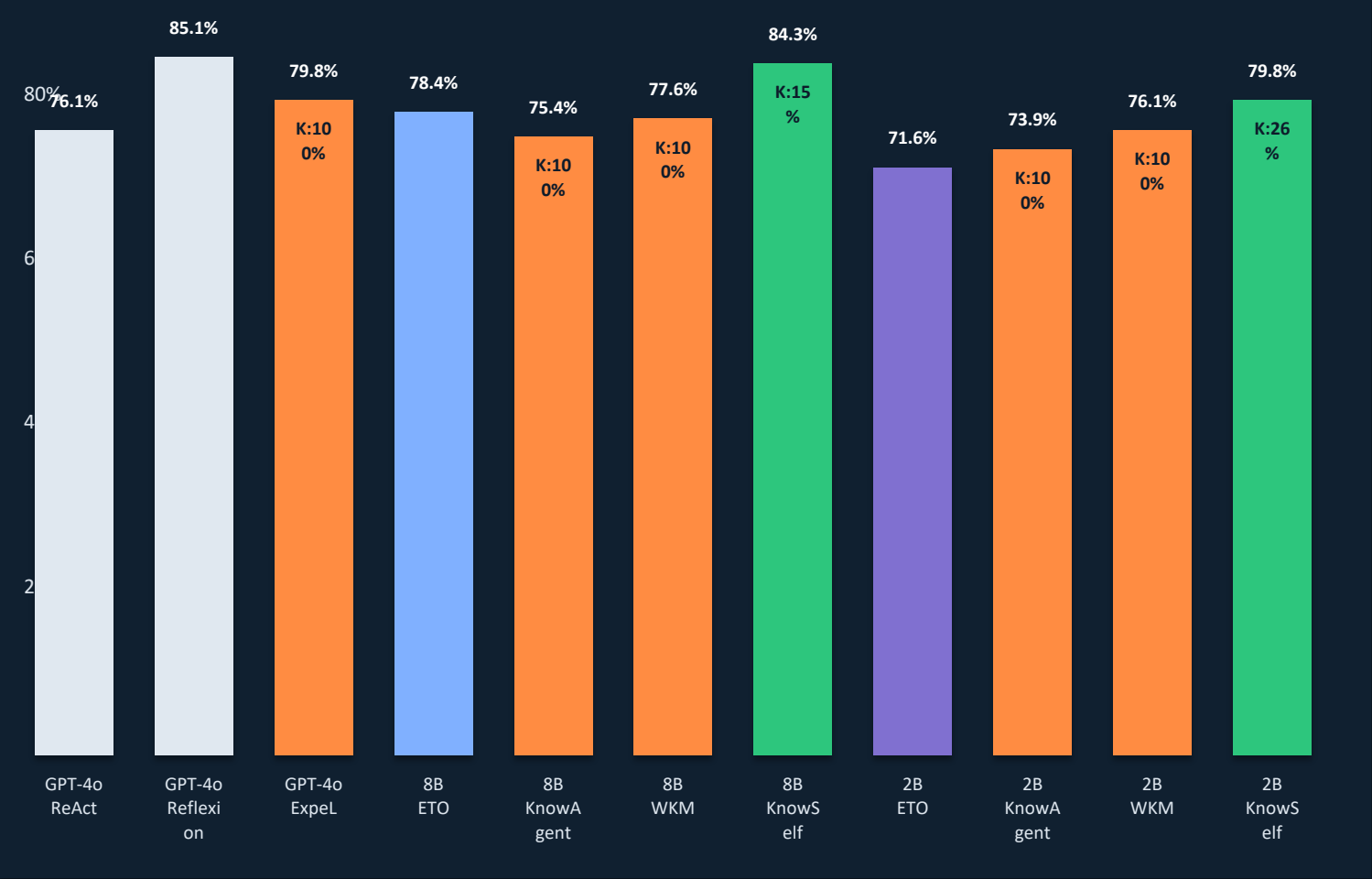
Success rates (%) across six ALFWorld task types; Know% = proportion of steps using external knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

★ KnowSelf (Llama-8B): 84.33% All — beats GPT-4o ExpeL (79.85%) using 100% knowledge, with only 15.01% knowledge queries

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage

~15–26% knowledge usage vs. 100% for KnowAgent/WKM/ExpeL — yet KnowSelf achieves best overall accuracy



Overall Accuracy (%) on ALFWorld

Green bar = KnowSelf
Orange bar = 100% KB methods
K:% = Know% used

Key Efficiency Insights

KnowSelf (Llama-8B)

84.33% accuracy
with only 15% knowledge queries

KnowSelf (Gemma-2B)

79.85% accuracy
with only 26% knowledge queries

Full-KB methods (ExpeL/KnowAgent/WKM)

100% knowledge usage
→ often lower accuracy

Gemma-2B KnowSelf

≈ GPT-4o ExpeL
79.85% vs 79.85%
at far lower model & KB cost

Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

Five findings that reframe how we should train and deploy LLM-based agents

01 Indiscriminate knowledge injection is wasteful — and harmful

Flood-irrigation methods (ExpeL, KnowAgent, WKM) apply external knowledge at every step. KnowSelf shows this hurts: selective use at 15–26% of steps yields higher accuracy.

02 Agents can learn situational self-awareness via special tokens

Lightweight heuristic labeling on self-explored trajectories — no human annotation. SFT + RPO teach the model to generate <fast>, <reflect>, or <knowledge> tokens accurately.

03 KnowSelf (Llama-8B) outperforms GPT-4o baselines

84.33% vs. 76.12% (GPT-4o ReAct) and 79.85% (GPT-4o ExpeL) — a smaller open model beats larger proprietary models through smarter resource allocation.

04 Only 15–26% knowledge usage needed — vs. 100% for competitors

6–7× reduction in knowledge queries without sacrificing accuracy. Significant inference-cost savings in deployment at scale.

05 The paradigm generalises across model scales

Even Gemma-2B (2B params) achieves 79.85%, matching GPT-4o ExpeL. Self-awareness is a training paradigm, not a model-size privilege.

Future Directions:

Extend to multi-agent settings • Apply to tool-use and code-generation agents • Explore adaptive KB construction • Formal metacognition benchmarks